

Problems: Categorical and Numeric Variables

Q1.

Consider the data: [red, blue, blue, red, green, grey]

- a. Represent these data as a matrix of one hot encoded variables
- b. Compute the sample proportions for each label

Consider the data:

$$X = [1, 3, 3, 6, 11]$$

- c. Compute and sketch the empirical CDF for this sample.

Q2.

We often want to take transformations of random variables, particularly when doing feature engineering.

- a. Suppose we have a random variable X and transform it as $Y = a + bX$, with $b > 0$. The distribution of X is given by $F_X(x)$. What is the distribution of Y , $F_Y(y)$? What is the density, $f_Y(y)$?

Suppose your sample is:

$$X = [1, 3, 4]$$

- b. What is the mean of the sample?
- c. Square all the values. What the mean of X^2 ?
- d. Compute the square of the mean from part b. Does this equal the mean of the square from part c?

Non-linear transformations change the moments/statistics of the data in a non-linear way. This is important to remember.

Q3.

The **exponential distribution** is

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases}$$

- a. Sketch a graph of the exponential distribution and density.
- b. What is the probability that $X \geq 3$? What is the probability that $X \leq 35$?
- c. Suppose we transform X by $Y = 1 + 3X$. What is the distribution of Y , $F_Y(y) = pr[Y \leq y]$? Provide a formula and sketch a graph in comparison to F_X . What is the probability that Y is less than 10?

Q4.

The **logistic distribution** is

$$F(x) = \frac{1}{1 + e^{-x}}$$

- a. Sketch a graph of the logistic distribution and density.
- b. What is the probability that $X \geq .8$? What is the probability that $X \leq .3$?
- c. Suppose we transform X by $Y = 2X - 1$. What is the distribution of Y ? Provide a formula and sketch a graph. What is the probability that Y is less than 0?

Q5.

The **median** is the value x for which the probability that X is above or below x is .5, or $F(\text{median}) = \frac{1}{2}$.

- a. What is the median of the exponential distribution?
- b. What is the median of the logistic distribution?

The **quantile function** is the inverse of the distribution function. The CDF answers the question, "What fraction u of the time is X below x ?" ($F(x) = u$) and the quantile function answers the question, "For what value x is X below x with probability u ?" ($F^{-1}(u) = x$)

- c. What is the quantile function of the exponential distribution?
- d. What is the quantile function of the logistic distribution?

Q6.

Load `./data/metabric.csv`.

- a. Make an ECDF plot of **Overall Survival (Months)**.
- b. Make an ECDF plot of **Overall Survival (Months)**, hued by **Chemotherapy**. Conditional on a patient receiving Chemotherapy, should we predict they will live a longer or shorter amount of time? Explain your answer clearly.
- c. Is chemotherapy an effective treatment? Explain your answer clearly.

Q1.

Consider the data: [red, blue, blue, red, green, grey]

- a. Represent these data as a matrix of one hot encoded variables
- b. Compute the sample proportions for each label

Consider the data:

$$X = [1, 3, 3, 6, 11]$$

- c. Compute and sketch the empirical CDF for this sample.

a.

iter	red	blue	green	gray
red	1	0	0	0
blue	0	1	0	0
Green	0	0	1	0
Gray	0	0	0	1

=> our example => data =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b.

$$\hat{p} = \frac{1}{n} \sum_{j=1}^n I_j$$

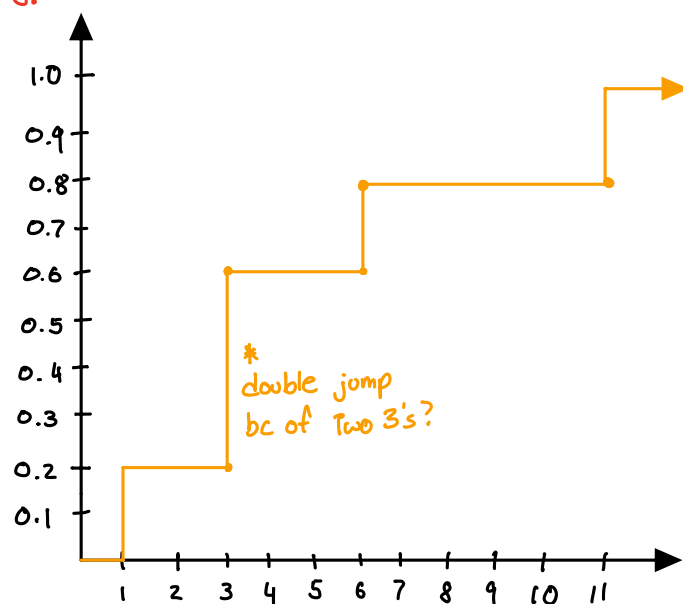
red | $\hat{p} = \frac{1}{6} \sum_{i=1}^6 2 = \frac{1}{3}$

blue | $\hat{p} = \frac{1}{6} \sum_{i=1}^6 2 = \frac{1}{3}$

green | $\hat{p} = \frac{1}{6} \sum_{i=1}^6 1 = \frac{1}{6}$

gray | $\hat{p} = \frac{1}{6} \sum_{i=1}^6 1 = \frac{1}{6}$

c.



$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n 1\{X_i \leq x\} = \frac{\# \text{ of observations}}{n}$$

$$1 \Rightarrow \frac{1}{5} = 0.2$$

$$3 \Rightarrow \frac{2}{5} = 0.4$$

$$6 \Rightarrow \frac{1}{5} = 0.2$$

$$11 \Rightarrow \frac{1}{5} = 0.2$$

Q2.

We often want to take transformations of random variables, particularly when doing feature engineering.

- a. Suppose we have a random variable X and transform it as $Y = a + bX$, with $b > 0$. The distribution of X is given by $F_X(x)$. What is the distribution of Y , $F_Y(y)$? What is the density, $f_Y(y)$?

Suppose your sample is:

$$X = [1, 3, 4]$$

F - the CDF; 'the probability of or below'

- b. What is the mean of the sample?
- c. Square all the values. What the mean of X^2 ?
- d. Compute the square of the mean from part b. Does this equal the mean of the square from part c?

Non-linear transformations change the moments/statistics of the data in a non-linear way. This is important to remember.

a. Probability density function

$$\hookrightarrow F(x) = \text{pr}[X \leq x] \quad F(y) = \text{pr}[a + bX \leq y]$$

$$F_Y(y) = F_X\left(\frac{y-a}{b}\right)$$

$$\hookrightarrow F(y) = \text{pr}[Y \leq y] \quad F(y) = \text{pr}\left[X \leq \frac{y-a}{b}\right] \quad f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X\left(\frac{y-a}{b}\right)$$

b. $A = [1, 3, 4]$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1+3+4}{3} = \frac{8}{3}$$

$$f_Y(y) = \frac{1}{b} f_X\left(\frac{y-a}{b}\right)$$

c. $\bar{X}^2 = \frac{1}{n} \sum x_i^2 = \frac{1+9+16}{3} = \frac{26}{3}$

d. $\left(\frac{8}{3}\right)^2 = \frac{64}{9} \approx 7.11 \neq \frac{26}{3}$

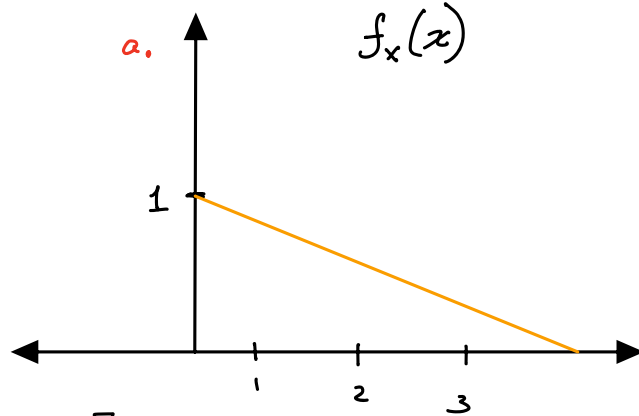
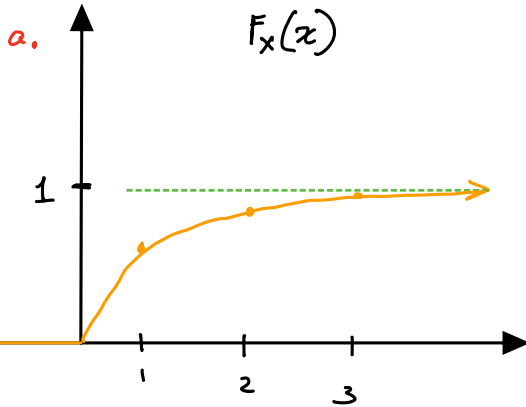
It does not equal to each other.
the square of means will always
be smaller than the mean of
squares.

Q3.

The exponential distribution is

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-x}, & x \geq 0 \end{cases} \quad f_x(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x \geq 0 \end{cases}$$

- a. Sketch a graph of the exponential distribution and density.
- b. What is the probability that $X \geq 3$? What is the probability that $X \leq 35$?
- c. Suppose we transform X by $Y = 1 + 3X$. What is the distribution of Y , $F_Y(y) = \text{pr}[Y \leq y]$? Provide a formula and sketch a graph in comparison to F_X . What is the probability that Y is less than 10?



b.

$$P[X \leq x] = 1 - P[X \geq x]$$

$$= 1 - (1 - e^{-x}) = 1 - 1 + e^{-x}$$

$$= e^{-x} = 0.0497 \approx 5\%$$

$$P[X \leq x] = F(35)$$

$$= 1 - e^{-35} \approx 100\%$$

c. PDF

$$F(x) = \text{pr}[X \leq x]$$

$$F(y) = \text{pr}[Y \leq y]$$

$$F(y) = \text{pr}[1 + 3X \leq y]$$

$$= \text{pr}\left[X \leq \frac{y-1}{3}\right]$$

$$\begin{aligned} x < 0 & \quad \frac{y-1}{3} < 0 \\ y & < 1 \end{aligned}$$

$$y < 1$$

$$x \geq 0$$

$$\frac{y-1}{3} \geq 0$$

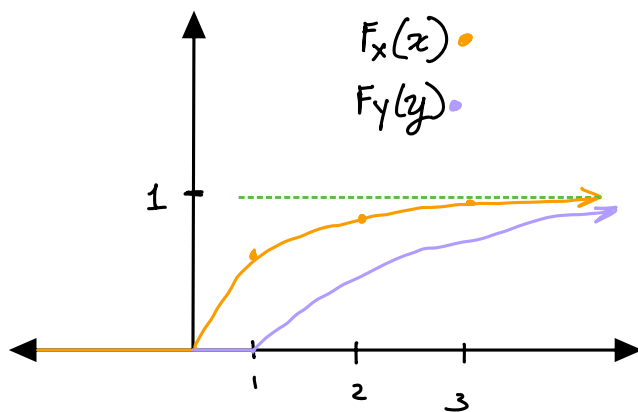
$$y \geq 1$$

$$y < 10$$

$$F(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-\left(\frac{y-1}{3}\right)} & y \geq 1 \end{cases}$$

$$F_Y(10) = 1 - e^{-\left(\frac{10-1}{3}\right)} = 1 - e^{-3}$$

$$F_Y(10) \approx 0.9502$$



Q4.

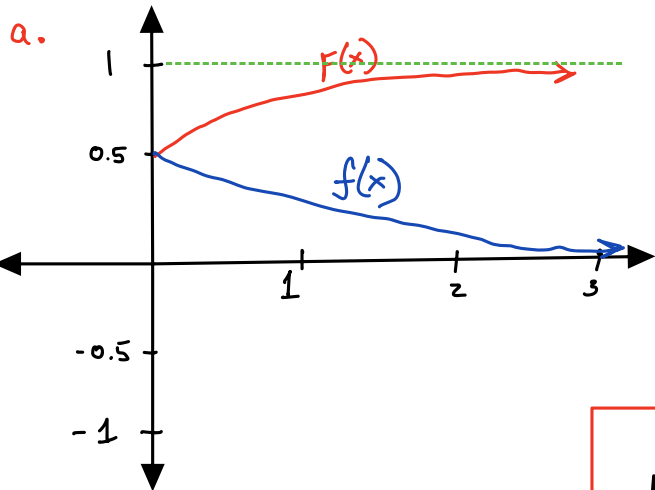
The logistic distribution is

$$F(x) = \frac{1}{1 + e^{-x}} (1 + e^{-x})^{-1}$$

$$f(x) = -1(1 + e^{-x})^{-1} \cdot e^{-x}$$

$$f(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

- a. Sketch a graph of the logistic distribution and density.
- b. What is the probability that $X \geq .8$? What is the probability that $X \leq .3$?
- c. Suppose we transform X by $Y = 2X - 1$. What is the distribution of Y ? Provide a formula and sketch a graph. What is the probability that Y is less than 0?



b. $x \geq 0.8$?
 we use the complement

$$P[X \leq x] = 1 - P[X \geq x]$$

$$= 1 - \left(\frac{1}{1 + e^{-0.8}} \right)$$

$$P[X \geq 0.8] = 0.310$$

$$x \leq 0.3? \quad P[X \leq 0.3] = \frac{1}{1 + e^{-0.3}}$$

$$P[X \leq 0.3] = 0.574$$

$$F(y) = \frac{1}{1 + e^{-\frac{(y+1)}{2}}}$$

Logistic =

$$F(x) = \frac{1}{1 + e^{-\frac{(x-\mu)}{\sigma}}}$$

c. $F(x) = \text{pr}[X \leq x]$

$$F(y) = \text{pr}[Y \leq y]$$

$$F(y) = \text{pr}[2X - 1 \leq y]$$

$$F(y) = \text{pr}\left[X \leq \frac{y+1}{2}\right]$$

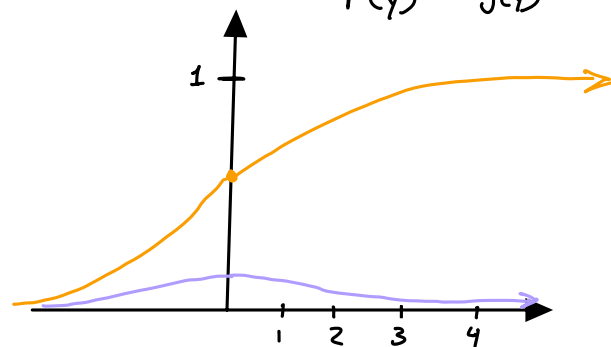
$$f(y) = F'(y) = \frac{1}{2} \frac{e^{-\frac{(y+1)}{2}}}{(1 + e^{-\frac{(y+1)}{2}})^2}$$

$$f(x) = \frac{e^{-\frac{(x-\mu)}{\sigma}}}{\sigma (1 + e^{-\frac{(x-\mu)}{\sigma}})^2}$$

$$\mu = -1, \quad \sigma = 2$$

$$F(y) \bullet f(y) \bullet$$

$$P(Y \leq 0) = F_Y(0) = F(y) = \frac{1}{1 + e^{-\frac{(0+1)}{2}}} = 0.622$$



Q5.

The **median** is the value x for which the probability that X is above or below x is .5, or $F(\text{median}) = \frac{1}{2}$.

- a. What is the median of the exponential distribution?
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The **quantile function** is the inverse of the distribution function. The CDF answers the question, "What fraction u of the time is X below x ?" ($F(x) = u$) and the quantile function answers the question, "For what value x is X below x with probability u ?" ($F^{-1}(u) = x$)

- c. What is the quantile function of the exponential distribution?
- d. What is the quantile function of the logistic distribution?

a. $F(x) = 1 - e^{-\lambda t} = 1/2$

$$e^{-\lambda t} = 1/2$$

$$\Rightarrow \boxed{\frac{\ln(2)}{\lambda} = t}$$

b. $F(x) = \frac{1}{1 + e^{\frac{-(x-\mu)}{\sigma}}} = 1/2 \Rightarrow 2 = 1 + e^{\frac{-(x-\mu)}{\sigma}}$

$$\ln 1 = \ln e^{\frac{-(x-\mu)}{\sigma}}$$

$$0 = \frac{-(x-\mu)}{\sigma}$$

$$\boxed{x = \mu}$$

c. $F(x) = 1 - e^{-\lambda x} = u$

$$-e^{-\lambda x} = u - 1$$

$$\ln(e^{-\lambda x}) = \ln(u - 1)$$

$$-\lambda x = \ln(u - 1)$$

$$\boxed{x = \frac{\ln(u - 1)}{-\lambda} = F^{-1}(u)}$$

d. $F(x) = \frac{1}{1 + e^{\frac{-(x-\mu)}{\sigma}}} = u$

$$\frac{1}{u} = 1 + e^{\frac{-(x-\mu)}{\sigma}}$$

$$\frac{1}{u} - 1 = e^{\frac{-(x-\mu)}{\sigma}}$$

$$\ln\left(\frac{1-u}{u}\right) = \frac{-(x-\mu)}{\sigma}$$

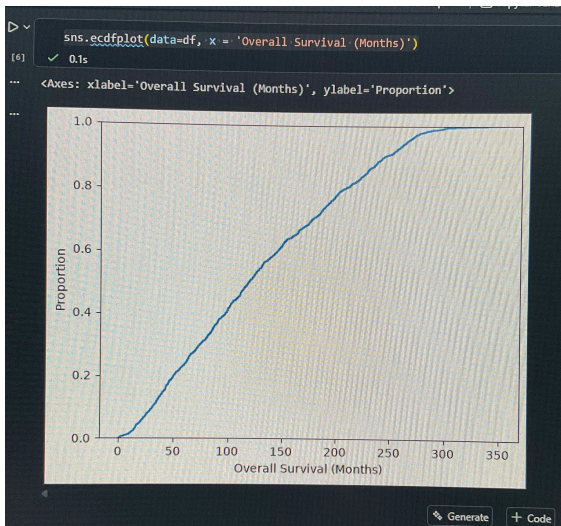
$$\boxed{\ln\left(\frac{1-u}{u}\right) \cdot \sigma + \mu = x = F^{-1}(u)}$$

Q6.

Load `./data/metabric.csv`.

- a. Make an ECDF plot of **Overall Survival (Months)**.
- b. Make an ECDF plot of **Overall Survival (Months)**, hued by **Chemotherapy**. Conditional on a patient receiving Chemotherapy, should we predict they will live a longer or shorter amount of time? Explain your answer clearly.
- c. Is chemotherapy an effective treatment? Explain your answer clearly.

a.



b.



c. Yes, I asked Claude to help me make a chart comparing Tumor stage with chemotherapy



For the most part the data supports my hypothesis that tumor stage and the proportion of those patients receiving chemo therapy scales linearly. However, stage 4 see a decrease when compared to stage 3. My explanation is that the cancer is too advanced for chemotherapy to be effective; instead the doctors prioritized comfort.

b. Based on that data we should predict that patients receiving chemotherapy live shorter than patients NOT on chemotherapy. However, we should not make this comparison as the two groups are **probably** not finding themselves in equal comparison. Chemo is extremely hard on the body; only prescribed when cancer is extremely advanced. So it stands to reason that this data of those patients that are undergoing chemotherapy have much more serious cases of cancer as opposed to the group that does not take chemotherapy.

